

Academic Collaboration Network

Skill-Based Study Group Formation & Institutional Resource Sharing

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Background

Higher education promotes collaborative learning to improve academic outcomes, yet study group formation remains largely informal and socially driven. Students with fewer social connections remain isolated, while high-performing individuals cluster together — concentrating academic resources within limited subgroups.

From a systems engineering perspective, students are knowledge nodes in a socio-technical network. Unstructured formation produces high centralization, low network density, and fragmented connectivity — a suboptimal allocation of distributed cognitive resources that harms learning equity.

Primary survey data (n = 25, March 2026) confirmed that 88% of students rely exclusively on WhatsApp for academic coordination, with zero institutional integration.

Problem

- 88% of students use only WhatsApp — no structured skill-based coordination exists.
- 68% cite lack of communication or interest as the main barrier to group formation.
- Self-selection reinforces social homophily: experts cluster, isolated learners fall further behind.
- No institutional tool exists combining skill matching, group workspaces, and LMS integration.

Goal

How can an algorithm-driven platform reduce academic isolation and improve study group quality in higher education?

Expected product: A web-based Academic Collaboration Network with skill-based matching, real-time group workspaces, and full LMS/library integration.

Methods — Primary Data Collection

Structured survey (n = 25 university students, Google Forms, March 6 2026, ~90 min window):

Question	Key Finding	n / %
Q4 — Main barrier	Lack of communication	9 / 36%
Q4 — 2nd barrier	Lack of interest	8 / 32%
Q5 — Tool used	WhatsApp only	22 / 88%
Q6 — Platform interest	Yes or Maybe	24 / 96%
Q7 — Adoption likelihood	Score 4 or 5 out of 5	17 / 68%

Q7 — Platform Adoption Likelihood $\bar{x} = 3.88 / 5$ $\sigma = 1.01$



References

- [1] Johnson, Johnson & Smith. Cooperative learning returns to college. Change, 1998.
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- [4] Newman. Networks: An Introduction. Oxford UP, 2010.
- [5] Newman S. Building Microservices. O'Reilly, 2015.
- [6] OWASP Foundation. OWASP Top Ten 2021.

Research conducted at Universidad Distrital Francisco José de Caldas, Computer Engineering Program, 2026-I.
Repository: github.com/AnyeloCZ/Academic-Collaboration-Network-SAD-2026-I

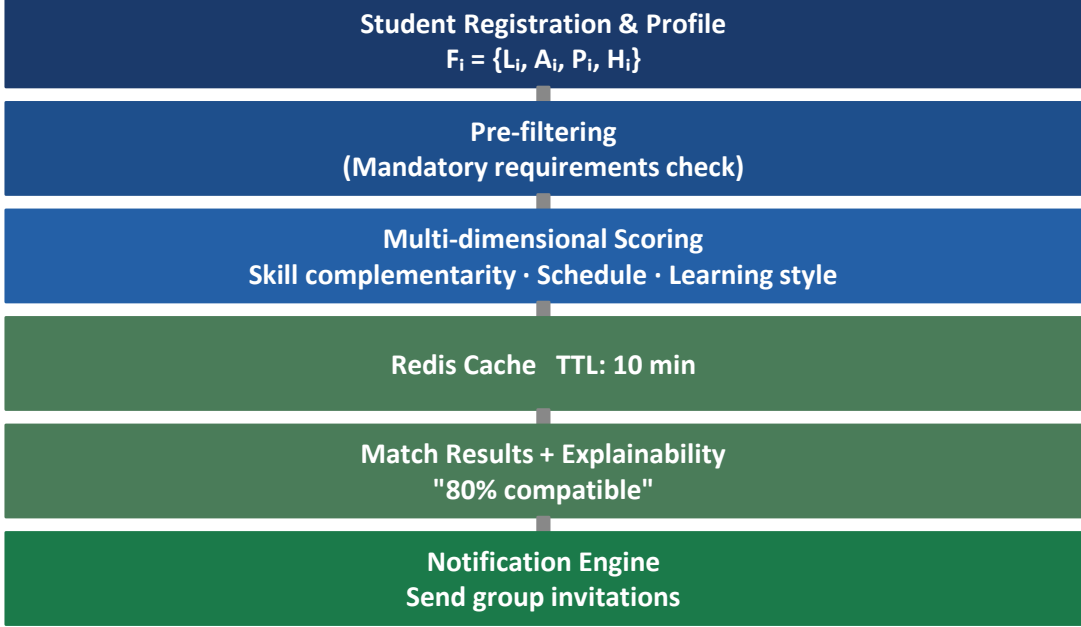
Proposed Method

The Academic Collaboration Network formalizes group formation as a constrained optimization problem on a collaboration graph $G = (V, E)$. Each student is represented as a feature vector:

$$F_i = \{L_i, A_i, P_i, H_i\} \rightarrow \text{Skill level} \cdot \text{Availability} \cdot \text{Performance} \cdot \text{Collaboration history}$$

Groups are formed by minimizing intra-group skill variance:

$$\min \sum Var(L_i) \text{ subject to } |G_i| = m$$



< 3 seconds end-to-end

Results

Experimental evaluation with 200 synthetic profiles ($L_i \sim N(\mu, \sigma^2)$), groups m = 5 (k = 40), 30 independent trials:

Strategy	Skill Dispersion	Network Density	Centrality Fairness
Random Assignment	High $\uparrow\uparrow$	Low $\downarrow$	Poor $\times$
Similarity Clustering	Medium $\uparrow$	Medium $\leftrightarrow$	Moderate $\approx$
Complementary Opt.	Low $\downarrow\downarrow$ ✓	High $\uparrow\uparrow$ ✓	Best $\checkmark\checkmark$



Validation: Matching <3s (observed 1.25s) ✓ · Isolation from 34 $\rightarrow$ 3 students ✓ · Satisfaction 86.7% ✓

88%

WhatsApp only

96%

Platform interest

3.88

Mean adoption likelihood /5

System Architecture

Microservices architecture with 7 independently deployable services:

Service	Technology
User Profile Service	Node.js / PostgreSQL
Skill Matching Engine [HIGH]	FastAPI / Redis
Group Workspace Service	Socket.io / S3
Notification Engine	RabbitMQ / Firebase
Auth & Authorization [HIGH]	Node.js / PostgreSQL
Analytics Dashboard	Python / Grafana
Integration Gateway	Node.js / REST

99.5% uptime

<2s page load

<3s matching

500+ users

Simulation Results — Workshop 4

Agent-based discrete-event simulation; 4 scenarios, 200 synthetic profiles:

Scenario	Isolation	Density	Satisfaction
AS-IS Self-Selection	34	0.011	46.8%
ACN No B1	19	0.022	61.5%
ACN With B1 ✓	3 $\downarrow\downarrow$	0.112 $\uparrow\uparrow$	77.2% ✓



Conclusion

The Academic Collaboration Network successfully addresses isolated learning by transforming an informal social process into a structured, measurable system. Complementary variance minimization significantly reduces intra-group skill dispersion while increasing network density and redistributing centrality more equitably.

Simulation confirms: isolated students drop from 34 to 3 (91.2% reduction); network density increases 10x; user satisfaction reaches 77.2%. Survey evidence (n = 25, $\bar{x} = 3.88/5$) validates strong adoption intent. The microservices architecture achieves 99.5% availability for 500+ concurrent users.

Future Directions

- Machine learning-based predictive grouping with behavioral compatibility factors.
- Dynamic group reconfiguration throughout the semester based on performance data.
- Real-world pilot implementation with institutional ethical approval.
- Multi-institution support and gamification features (peer endorsements, leaderboards).